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# Apolipoprotein A-I and the molecular variant apoA-I$_{Milano}$: Evaluation of the antiatherogenic effects in knock-in mouse model

Cinzia Parolini [a,*], Giulia Chiesa [a], Elaine Gong [b], Silvia Caligari [a], Miriam M. Cortese [a], Teiichiro Koga [b], Trudy M. Forte [b], Edward M. Rubin [b]

[a] *Department of Pharmacological Sciences, University of Milan, 20133 Milan, Italy*
[b] *Genome Sciences Department, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA*

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**Abstract**

No evidence of premature vascular disease is found in apolipoprotein A-I$_{Milano}$ (apoA-I$_M$) human carriers, despite very low high density lipoprotein (HDL) cholesterol levels. Whether apoA-I$_M$ may impart a "gain of function" in atherosclerosis protection compared to wild-type apoA-I is hotly debated. To address this question, knock-in mice expressing human apoA-I or apoA-I$_M$ were crossed with atherosclerosis-susceptible mice expressing the human apoB/A-II transgene (h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$_M^{Hu/Hu}$). On a chow diet, h-B/A-II/A-I$_M^{Hu/Hu}$ mice were characterized by low HDL cholesterol levels compared to h-B/A-II/A-I$^{Hu/Hu}$ mice ($35.65 \pm 8.00$ mg/dl versus $58.09 \pm 13.50$ mg/dl, respectively; $p < 0.005$). Gender differences in response to high fat diet were observed in both h-B/A-II/A-I$_M^{Hu/Hu}$ and h-B/A-II/A-I$^{Hu/Hu}$ lines. h-B/A-II/A-I$_M^{Hu/Hu}$ females had higher total cholesterol levels compared to h-B/A-II/A-I$^{Hu/Hu}$ females ($895.08 \pm 183.07$ mg/dl versus $544.43 \pm 116.42$ mg/dl; $p < 0.05$) and developed larger atherosclerotic lesions ($148{,}260 \pm 78{,}924\ \mu m^2$ versus $54{,}132 \pm 43{,}204\ \mu m^2$, respectively; $p < 0.05$). On the contrary, no difference in mean lesion area was found between h-B/A-II/A-I$_M^{Hu/Hu}$ and h-B/A-II/A-I$^{Hu/Hu}$ males ($19{,}779 \pm 6098\ \mu m^2$ versus $15{,}706 \pm 13{,}095\ \mu m^2$; $p = 0.685$). Our data suggest that, in the atherosclerosis-susceptible human apoB/A-II mouse model, expression of the human *apoA-I$_M$* gene does not have protective advantage over that of the *apoA-I* gene.
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*Keywords:* Apolipoprotein A-I; HDL cholesterol; Gene knock-in; Mouse model; Atherosclerosis

## 1. Introduction

Apolipoprotein A-I$_{Milano}$ (apoA-I$_M$), wherein a cysteine is substituted for an arginine at position 173, is the first described mutant of apoA-I [1]. Despite a high-risk lipid profile, characterized by a severe hypoalphalipoproteinemia, normal to elevated plasma low density lipoprotein (LDL) cholesterol, and a moderate hypertriglyceridemia [2], no evidence of premature vascular disease is found in apoA-I$_M$ human carriers [3]. This suggests that the mutant possesses special features, which confer a high degree of protection against vascular disease [4]. Indeed, apoA-I$_M$ displays better antioxidant properties compared to human wild-type apoA-I [5] and it is more effective in promoting cholesterol efflux from cells, the first step in reverse cholesterol transport [6]. In addition, intravascular infusions of recombinant apoA-I$_M$ have reduced atherosclerosis and restenosis development in mice [7] and rabbits [8–10]. These data led to the successful use of recombinant apoA-I$_M$ as a potential therapeutic approach for atherosclerosis in humans [11]. Recently, in our laboratory a gene knock-in (k-in) strategy was used to generate two animal models that differ only in the biochemical nature of the apoA-I, i.e. carrying in one case the human wild-type *apoA-I* gene (A-I k-in) and, in the other, the *apoA-I$_M$* gene (A-I$_M$ k-in) [12]. The phenotype of the A-I$_M$ k-in mouse line reflects many of the features observed in the apoA-I$_M$ carriers.

In order to study these two forms of apoA-I with regard to their effects on atherosclerosis, the two k-in lines were crossed with human apoB/A-II expressing mice to

* Corresponding author. Tel.: +39 02 50318328; fax: +39 02 50318284.
*E-mail address:* cinzia.parolini@unimi.it (C. Parolini).

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obtain mouse lines with a genetic background susceptible to diet-induced atherosclerosis (h-B/A-II/A-$I_M^{Hu/Hu}$ and h-B/A-II/A-$I^{Hu/Hu}$). The h-B/A-II/A-$I_M^{Hu/Hu}$ line had a high-risk pre-diet lipid profile compared to h-B/A-II/A-$I^{Hu/Hu}$ mice. After the atherogenic diet, h-B/A-II/A-$I_M^{Hu/Hu}$ females displayed higher total cholesterol levels and developed larger atherosclerotic lesions compared to all the other groups analyzed. No difference in the mean lesion area was observed between h-B/A-II/A-$I_M^{Hu/Hu}$ and h-B/A-II/A-$I^{Hu/Hu}$ male mice.

## 2. Materials and methods

### 2.1. Transgenic mice

The human apoB/human apoA-II (h-B/A-II) transgenic mouse line used for the study was created in an FVB background. The transgene in these mice was derived from the fusion of two constructs previously microinjected into mouse oocytes to obtain human apoB [13] and human apoA-II transgenic mice [14]. Briefly, an apoB-containing P1 clone was isolated by PCR using primers to exon 18 of the *apoB* gene as described [13]. The human *apoA-II* gene (A-IIg) fused to the human apoA-I promoter (A-Ip) [14] was inserted into a NotI/SalI digested pBC vector (pBC/A-Ip/A-IIg). To obtain the h-B/A-II transgene, the pBC/A-Ip/A-IIg was digested with BssHII and inserted into a BssHII digested apoB P1 phagemid. DNA was gently treated with GeneClean before injection into inbred FVB zygotes as described [13]. One of the two founders obtained was crossed with FVB mice in order to establish the new mouse line.

Previously generated homozygous k-in lines expressing human (Hu) apoA-I (A-$I^{Hu/Hu}$) or human apoA-$I_M$ (A-$I_M^{Hu/Hu}$) mice [12] were backcrossed for three generations in an FVB background. Subsequently, hemizygous h-B/A-II mice were crossed with A-$I^{Hu/Hu}$ or A-$I_M^{Hu/Hu}$ to generate h-B/A-II/A-$I^{Hu/+}$ and h-B/A-II/A-$I_M^{Hu/+}$ mice, respectively. h-B/A-II/A-$I^{Hu/+}$ and A-$I^{Hu/+}$ animals were intercrossed to create h-B/A-II/A-$I^{Hu/Hu}$ and h-B/A-II/A-$I^{+/+}$ mice for experiments. The same strategy was adopted to generate h-B/A-II/A-$I_M^{Hu/Hu}$ and h-B/A-II/A-$I^{+/+}$ animals. The h-B/A-II/A-$I^{+/+}$ littermates of the two transgenic lines were compared for the lipid/lipoprotein phenotype, and no significant differences were observed.

The transgenic mouse lines used in this study were h-B/A-II/A-$I^{Hu/Hu}$, h-B/A-II/A-$I_M^{Hu/Hu}$, and h-B/A-II/A-$I^{+/+}$, of both sexes. Procedures involving animals and their care were conducted in accordance with institutional guidelines that are in compliance with national (D.L. No. 116, G.U. Suppl. 40, February 18, 1992, Circolare No. 8, G.U. luglio 1994) and international laws and policies (EEC Council Directive 86/609, OJL 358, 1, December 12, 1987) [15].

### 2.2. Screening of transgenic lines

h-B/A-II transgenic mice were identified by PCR with specific primers for human apoB (Fw/apoB5′: 5′-GTGCCTCCCTCTTCAGTTACC-3′; Rw/apoB5′: 5′-ACGTCACAGCCAATAATGAGC-3′) and for human apoA-II (Fw/apoA-II: 5′-TGACTCTATTCCTACCTAGGGGC-3′; Rw/apoA-II: 5′-GTTCCAAGTTCCACGAAATAGC-3′). Human apoB and apoA-II expression was assessed by Western blot analysis using specific polyclonal antibodies (LTA, Italy). Mouse apoA-I (Fw/m-apoA-I: 5′-TTGGATATCTCGCACCTTTAGC-3′; Rw/m-apoA-I: 5′-AGGGAAGAGAACAGTGGGAAT-3′) and human apoAI/apoA-$I_M$ specific primers (Fw/apoA-I: 5′-AGCGCTCTCGAGGAGTACAC-3′; Rw/neo: 5′-CCACCAAAGAACGGAGCC-3′) were used to screen by PCR the h-B/A-II/A-$I^{Hu/Hu}$, h-B/A-II/A-$I_M^{Hu/Hu}$, and h-B/A-II/A-$I^{+/+}$ mouse lines.

### 2.3. Diets, apolipoprotein and lipoprotein analyses

Ten male and 10 female mice for each genotype were fed a mouse chow diet (4RF 21, Mucedola, Milano) until 8–10 weeks of age. Before starting the atherogenic diet, the transgenic lines and FVB control animals were fasted overnight and blood was collected from the retro-orbital plexus into tubes containing 0.1% (w/v) EDTA for lipid-apolipoprotein analysis. h-B/A-II/A-$I^{Hu/Hu}$, h-B/A-II/A-$I_M^{Hu/Hu}$, and h-B/A-II/A-$I^{+/+}$ were then placed on an atherogenic diet consisting of 1.25% cholesterol, 0.5% cholic acid, and 15% fat [16]. Mice were maintained on the diet for 18 weeks; then, blood was collected after an overnight fast for lipid-apolipoprotein profile and all the animals were sacrificed for histological analysis. Plasma total cholesterol was measured by enzymatic methods (Hoffmann La Roche, Basel; Switzerland; Boehringer Mannheim, Mannheim, Germany) [17]. Triglyceride (TG) concentrations were corrected for the free glycerol present in plasma as described (Sigma–Aldrich, Germany) [14]. HDL cholesterol levels were measured after precipitation of apoB-containing lipoproteins with PEG 8000 (20%, w/v) in 0.2 M glycine (pH 10) [14]. Human apolipoprotein concentrations were determined by immunoturbidimetric assays as previously described [18], using polyclonal antibodies specific for human apoA-I/A-$I_M$, human apoB and human apoA-II (LTA, Milano).

Pooled plasma samples of five animals from each transgenic line were applied onto a Superose 6 10/30 column (Pharmacia) as previously described [18]. Twenty fractions (0.8 ml each) were collected, and cholesterol and TG contents were measured by the enzymatic methods described above.

### 2.4. Atherosclerosis lesion area

After 18 weeks of feeding the atherogenic diet, mice were sacrificed, and their hearts were collected. Aortic sectioning, lipid staining, and lesion scoring were performed as previously described with some modification [19]. Briefly, the heart and attached aorta were first perfused with phosphate-buffered saline, then perfused with diluted tissue embedding medium (Tissue-Tek O.C.T., Miles Inc., Elkhart, in $H_2O$, 1:1)

and quickly frozen in O.C.T. 10-μm-thick sections were collected, starting with the first and most proximal section of the aorta where the aorta becomes round and the aortic valves distinct. Sections were stained with oil-red O and hematoxylin; the lesion area was determined by measuring the oil-red O stained lesions using a calibrated eyepiece at 200× magnifications. The mean lesion area/section/animal was determined for each individual animal.

### 2.5. Statistical analysis

Differences among groups were evaluated by two-way ANOVA or by multivariate analysis of variances (repeated measures).

## 3. Results

### 3.1. Generation of transgenic lines

Two founder mice positive for both human apoB and human apoA-II sequences were obtained by microinjection of fertilized eggs. One of the two founders was selected to establish the transgenic line. All h-B/A-II mice used in the study were obtained by crossing h-B/A-II positive mice and FVB control animals; therefore they were all hemizygous for the transgene. h-B/A-II mice were used to generate, by multiple crosses, h-B/A-II/A-$I^{Hu/Hu}$, h-B/A-II/A-$I_M{}^{Hu/Hu}$, and h-B/A-II/A$I^{+/+}$ mice as described in Section 2.

### 3.2. Plasma lipids and apolipoproteins in mice fed normal chow

#### 3.2.1. Lipid and apolipoprotein levels

On a normal chow diet there was no difference between females and males, in apolipoprotein and lipid plasma levels, in each mouse line analyzed. Table 1 shows the combined data for mice of both genders.

As expected, in control FVB mice more than 80% of cholesterol was present in the HDL fraction [13]. Expression of human apoB/A-II transgene (h-B/A-II/A-$I^{+/+}$) in the FVB strain resulted in a significant reduction of HDL levels with no changes in total cholesterol, and a consequent elevation of non-HDL cholesterol (38.18 ± 18.03 mg/dl versus 17.02 ± 9.63 mg/dl; $p < 0.05$). Plasma triglyceride levels in h-B/A-II/A-$I^{+/+}$ mice were higher compared to those measured in controls, but did not attain statistical significance.

The expression of human apoA-I or apoA-$I_M$ in a human apoB/A-II transgenic background (h-B/A-II/A-$I^{Hu/Hu}$ and h-B/A-II/A-$I_M{}^{Hu/Hu}$) did not modify human apoB and apoA-II plasma levels (see Table 1). No difference was also observed in apoA-I concentrations between h-B/A-II/A-$I_M{}^{Hu/Hu}$ and h-B/A-II/A-$I^{Hu/Hu}$ mice. The lipid profile of h-B/A-II/A-$I^{Hu/Hu}$ mice was similar to that of h-B/A-II/A-$I^{+/+}$ mice. In contrast, h-B/A-II/A-$I_M{}^{Hu/Hu}$ transgenic line had significantly lower plasma HDL cholesterol levels than those observed in both h-B/A-II/A-$I^{+/+}$ and h-B/A-II/A-$I^{Hu/Hu}$ mice, with no variation in total cholesterol. As a consequence, non-HDL cholesterol levels in h-B/A-II/A-$I_M{}^{Hu/Hu}$ mice were significantly higher compared to those measured in h-B/A-II/A-$I^{Hu/Hu}$ and h-B/A-II/A-$I^{+/+}$ lines (59.84 ± 19.03 mg/dl versus 34.79 ± 8.33 mg/dl in h-B/A-II/A-$I^{Hu/Hu}$ and 38.18 ± 18.03 mg/dl in h-B/A-II/A-$I^{+/+}$ mice; $p < 0.05$). In addition, h-B/A-II/A-$I_M{}^{Hu/Hu}$ mice displayed elevated triglyceride levels compared to h-B/A-II/A-$I^{+/+}$ transgenic line.

#### 3.2.2. FPLC lipoprotein analysis

Pooled plasma samples from each transgenic line (three females and two males) were separated by gel filtration chromatography on a Superose 6B column for the determination of cholesterol and triglyceride distribution (Fig. 1). In h-B/A-II/A-$I^{+/+}$ mice the cholesterol was mainly found in the HDL fraction, but a significant cholesterol accumulation was observed in LDL particles, consistent with previously published data on human apoB expressing mice [20]. The cholesterol distribution in h-B/A-II/A-$I^{Hu/Hu}$ mice was nearly identical to that of h-B/A-II/A-$I^{+/+}$ mice. The cholesterol profile of h-B/A-II/A-$I_M{}^{Hu/Hu}$ mice was different from that of the other two mouse lines: cholesterol preferentially accumulated in the LDL fraction and the peak corresponding to the HDL fraction was markedly lower (Fig. 1A).

No differences were observed in the TG distribution between h-B/A-II/A-$I^{Hu/Hu}$ and h-B/A-II/A-$I^{+/+}$ mice. FPLC analysis of h-B/A-II/A-$I_M{}^{Hu/Hu}$ mice confirmed the increase of plasma TG measured in whole plasma (Table 1) and showed that TG mostly accumulated in the VLDL and LDL fractions (Fig. 1B).

Table 1
Plasma lipid profiles and human apolipoprotein concentrations in mice (10 males + 10 females) fed normal chow

| Mice | TC (mg/dl) | HDL-C (mg/dl) | TG (mg/dl) | Hu-A-I or Hu-A-$I_M$ (mg/dl) | Hu-A-II (mg/dl) | Hu-B (mg/dl) |
|---|---|---|---|---|---|---|
| h-B/A-II/A-$I_M{}^{Hu/Hu}$ | 95.49 ± 20.19 | 35.65 ± 8.00[a,b] | 79.25 ± 30.00[d] | 109.66 ± 30.79 | 2.05 ± 0.89 | 47.04 ± 18.86 |
| h-B/A-II/A-$I^{Hu/Hu}$ | 96.59 ± 14.56 | 58.09 ± 13.50[c] | 61.44 ± 28.93 | 112.75 ± 24.29 | 2.31 ± 1.03 | 40.95 ± 24.89 |
| h-B/A-II/A-$I^{+/+}$ | 102.13 ± 15.61 | 63.95 ± 14.24[c] | 43.56 ± 22.87 | – | 1.85 ± 0.71 | 43.16 ± 14.88 |
| Controls | 101.16 ± 9.47 | 83.89 ± 10.47 | 28.25 ± 11.14 | – | – | – |

Results are expressed as means ± S.D.; TC, total cholesterol.
[a] $p < 0.005$ vs. h-B/A-II/A-$I^{Hu/Hu}$ mice.
[b] $p < 0.001$ vs. h-B/A-II/A-$I^{+/+}$ and control mice.
[c] $p < 0.05$ vs. control mice.
[d] $p < 0.005$ vs. h-B/A-II/A-$I^{+/+}$ and control mice.

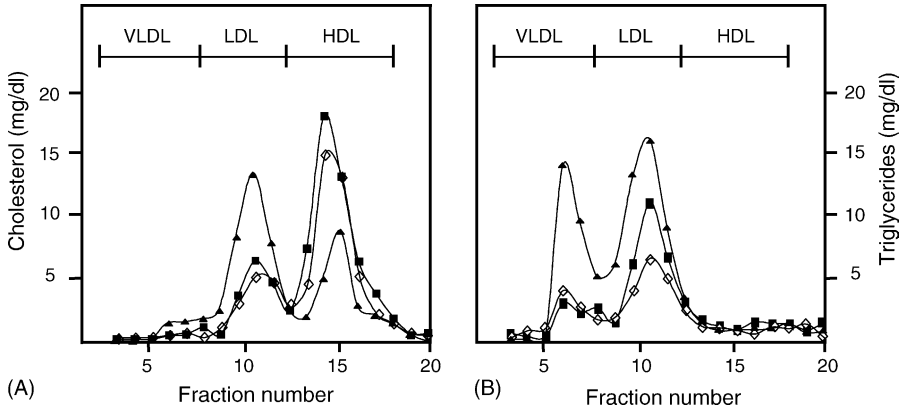



Fig. 1. FPLC profile of plasma cholesterol (A) and triglycerides (B) in h-B/A-II/A-I$^{+/+}$(open diamonds), h-B/A-II/A-I$^{Hu/Hu}$ (filled squares) and h-B/A-II/A-I$_M$$^{Hu/Hu}$ (filled triangles) transgenic lines. Pooled plasma samples of five mice (three females and two males) from each line were subjected to FPLC fractionation analysis.

### 3.3. Lipid and apolipoprotein response to the atherogenic diet

The same transgenic animals analyzed for lipid profiles on chow diet were fed a high fat, high cholesterol diet for 18 weeks. Females and males responded differently to the atherogenic diet, therefore lipid and apolipoprotein data are shown in separate tables (Tables 2 and 3, respectively).

#### 3.3.1. Lipid and apolipoprotein levels in females

After the high fat diet all female mouse lines had a marked increase of total cholesterol and human apoB plasma levels, and a reduction in TG plasma concentrations compared to chow diet values (Table 2). The observed decrease in triglyceride levels is characteristic of the pro-atherogenic, cholate-containing diet used in this study [21,22]. Human apoA-II levels were reduced below the sensitivity of the quantification method (<0.5 mg/dl, data not shown). In contrast to other mouse lines, human apoA-I expressing females had significantly reduced HDL cholesterol concentrations (Table 2). Comparison among groups after the atherogenic diet indicated that total cholesterol levels were higher in the h-B/A-II/A-I$_M$$^{Hu/Hu}$ line compared to both h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$^{+/+}$ female mice.

#### 3.3.2. Lipid and apolipoprotein levels in males

Similarly to females, in male h-B/A-II/A-I$^{+/+}$ transgenic mice the atherogenic diet determined a marked increase in total cholesterol and human apoB plasma levels and a decrease in TG concentrations, compared to chow diet values (Table 3). In h-B/A-II/A-I$^{Hu/Hu}$ mice the pro-atherogenic diet led to a moderate, but significant, total cholesterol elevation and to a reduction of TG levels. No significant variations were observed for human apoB plasma levels compared to chow diet concentrations. In h-B/A-II/A-I$_M$$^{Hu/Hu}$ male mice besides the same changes observed in h-B/A-II/A-I$^{+/+}$ mice, i.e. total cholesterol and apoB elevation and TG reduction, a significant rise in HDL cholesterol levels was detected (Table 3). As in females, human apoA-II concentrations were not detectable in all mouse lines analyzed (<0.5 mg/dl, data not shown). Comparison of the lipid-apolipoprotein profile among the three mouse lines after the atherogenic diet (Table 3) showed significantly lower levels of total cholesterol and human apoB in h-B/A-II/A-I$^{Hu/Hu}$ mice compared to the h-B/A-II/A-I$^{+/+}$ line.

#### 3.3.3. Comparison between females and males for lipid and apolipoprotein levels

Statistical comparisons between female and male mice for plasma lipid-apolipoprotein levels after the atherogenic diet did not show differences in the h-B/A-II/A-I$^{+/+}$ line. In contrast, in both h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$_M$$^{Hu/Hu}$ females, total cholesterol concentrations were higher compared to those measured in h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$_M$$^{Hu/Hu}$ males ($p < 0.05$ and $< 0.005$, respectively). Furthermore, h-B/A-II/A-I$_M$$^{Hu/Hu}$ females were characterized by higher human apoB ($p < 0.005$) and lower HDL cholesterol levels ($p < 0.05$) compared to h-B/A-II/A-I$_M$$^{Hu/Hu}$ males.

#### 3.3.4. FPLC lipoprotein analysis

Since the atherogenic diet differently affected the lipid and apolipoprotein levels in h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$_M$$^{Hu/Hu}$ female and male mice, separate pools of plasma samples obtained from five males or five females from each transgenic line were fractionated by FPLC analysis. In female mice, cholesterol and TG distribution among the lipoprotein fractions are illustrated in Fig. 2. In all mouse lines, the majority of cholesterol was detected in VLDL and LDL particles (Fig. 2A). As expected from the lipid data shown in Table 2, cholesterol accumulation was more elevated in h-B/A-II/A-I$_M$$^{Hu/Hu}$ mice compared to h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$^{+/+}$ trangenic lines (Fig. 2A). The TG distribution (Fig. 2B) showed no dramatic differences among the three mouse lines, but displayed moderate elevation of the TG content in the h-B/A-II/A-I$_M$$^{Hu/Hu}$ mice, as also indicated in Table 2.

Table 2
Plasma lipid profiles and human apolipoprotein concentration in female transgenic mice on chow diet (chow) and on atherogenic diet (Ath)

| Genotype | Chow | | | | | Ath | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | TC (mg/dl) | HDL-C (mg/dl) | TG (mg/dl) | Hu-A-I or Hu-A-$I_M$ (mg/dl) | Hu-B (mg/dl) | TC (mg/dl) | HDL-C (mg/dl) | TG (mg/dl) | Hu-A-I or Hu-A-$I_M$ (mg/dl) | Hu-B (mg/dl) |
| h-B/A-II/A-$I_M^{Hu/Hu}$ | 99.79 ± 26.83 | 38.31 ± 7.49 | 85.84 ± 39.52 | 116.97 ± 35.75 | 55.58 ± 23.39 | 895.08 ± 183.07 | 42.61 ± 9.12 | 29.64 ± 10.31 | 101.64 ± 34.59 | 203.22 ± 32.46 |
| h-B/A-II/A-$I^{Hu/Hu}$ | 99.55 ± 15.64 | 65.17 ± 14.39 | 55.44 ± 35.59 | 93.38 ± 37.04 | 44.08 ± 35.31 | 544.43 ± 116.42 | 32.14 ± 7.69 | 12.65 ± 1.70 | 71.82 ± 39.18 | 139.85 ± 60.21 |
| h-B/A-II/A-$I^{+/+}$ | 95.00 ± 20.29 | 66.63 ± 12.42 | 46.8 ± 14.88 | – | 50.08 ± 16.87 | 588.91 ± 197.31 | 55.41 ± 19.65 | 14.29 ± 8.10 | – | 133.60 ± 11.76 |
| *P* value | | | | | | | | | | |
| A-$I_M^{Hu/Hu}$ vs. A-$I^{Hu/Hu}$ | n.s. | 0.0134 | n.s. | n.s. | n.s. | 0.0293 | n.s. | n.s. | n.s. | n.s. |
| A-$I_M^{Hu/Hu}$ vs. A-$I^{+/+}$ | n.s. | 0.0067 | n.s. | n.s. | n.s. | 0.0429 | n.s. | n.s. | n.s. | n.s. |
| A-$I^{Hu/Hu}$ vs. A-$I^{+/+}$ | n.s. | n.s. | n.s. | – | n.s. | n.s. | n.s. | n.s. | – | n.s. |
| Chow vs. ath (A-$I_M^{Hu/Hu}$) | | | | | | 0.0003 | n.s. | 0.0293 | n.s. | 0.0003 |
| Chow vs. ath (A-$I^{Hu/Hu)}$ | | | | | | 0.0005 | 0.0069 | 0.0412 | *n.s.* | 0.0334 |
| Chow vs. ath (A-$I^{+/+)}$ | | | | | | 0.0015 | n.s. | 0.0011 | – | 0.0005 |

Results are expressed as means ± S.D.; TC, total cholesterol.

Table 3
Plasma lipid profiles and human apolipoprotein concentration in male transgenic mice on chow diet (chow) and on atherogenic diet (Ath)

| Genotype | Chow | | | | | Ath | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | TC (mg/dl) | HDL-C (mg/dl) | TG (mg/dl) | Hu-A-I or Hu-A-$I_M$ (mg/dl) | Hu-B (mg/dl) | TC (mg/dl) | HDL-C (mg/dl) | TG (mg/dl) | Hu-A-I or Hu-A-$I_M$ (mg/dl) | Hu-B (mg/dl) |
| h-B/A-II/A-$I_M^{Hu/Hu}$ | 91.19 ± 12.28 | 32.99 ± 8.39 | 55.70 ± 20.06 | 96.16 ± 19.15 | 38.50 ± 8.43 | 374.72 ± 84.13 | 60.57 ± 3.17 | 17.06 ± 10.60 | 128.49 ± 39.87 | 93.66 ± 20.13 |
| h-B/A-II/A-$I^{Hu/Hu}$ | 93.63 ± 15.06 | 55.49 ± 9.23 | 67.44 ± 20.07 | 108.3 ± 28.51 | 37.83 ± 13.13 | 271.32 ± 105.07 | 48.33 ± 17.60 | 17.07 ± 7.34 | 85.83 ± 32.38 | 63.75 ± 18.00 |
| h-B/A-II/A-$I^{+/+}$ | 109.26 ± 3.06 | 60.27 ± 17.97 | 37.32 ± 10.31 | – | 36.23 ± 9.64 | 464.76 ± 48.38 | 46.93 ± 13.83 | 7.27 ± 5.42 | – | 122.05 ± 31.83 |
| *P* value | | | | | | | | | | |
| A-$I_M^{Hu/Hu}$ vs. A-$I^{Hu/Hu}$ | n.s. | 0.0251 | n.s. | n.s. | n.s. | n.s. | n.s. | n.s. | n.s. | *n.s.* |
| A-$I_M^{Hu/Hu}$ vs. A-$I^{+/+}$ | n.s. | 0.0184 | n.s. | n.s. | n.s. | n.s. | n.s. | n.s. | n.s. | *n.s.* |
| A-$I^{Hu/Hu}$ vs. A-$I^{+/+}$ | n.s. | n.s. | n.s. | – | n.s. | 0.0224 | n.s. | n.s. | – | 0.019 |
| Chow vs. ath (A-$I_M^{Hu/Hu}$) | | | | | | 0.0004 | 0.002 | 0.0176 | n.s. | 0.001 |
| Chow vs. ath (A-$I^{Hu/Hu}$) | | | | | | 0.0156 | n.s. | 0.0038 | n.s. | *n.s.* |
| Chow vs. ath (A-$I^{+/+}$) | | | | | | 0.0002 | n.s. | 0.0024 | – | 0.0009 |

Results are expressed as means ± S.D.; TC, total cholesterol.

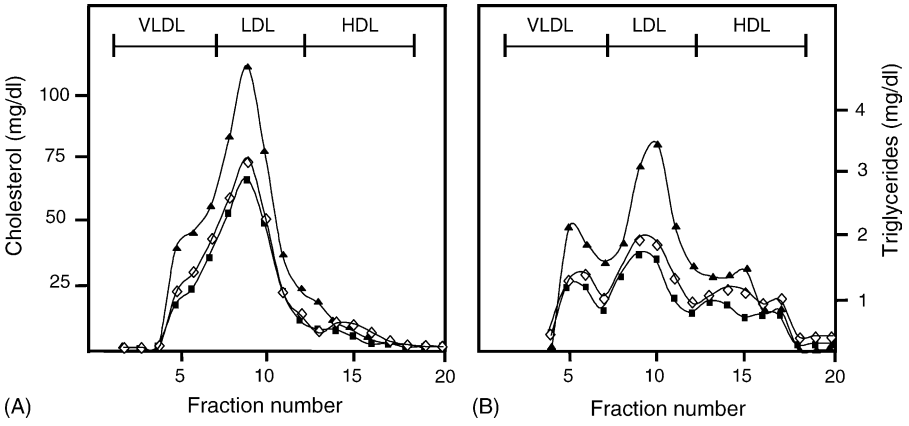



Fig. 2. FPLC profile of plasma cholesterol (A) and triglycerides (B) in h-B/A-II/A-I$^{+/+}$(open diamonds), h-B/A-II/A-I$^{Hu/Hu}$ (filled squares) and h-B/A-II/A-I$_M$$^{Hu/Hu}$ (filled triangles) transgenic females after 18 weeks of atherogenic diet. Pooled plasma samples of five mice from each line were applied to a Superose 6B column.

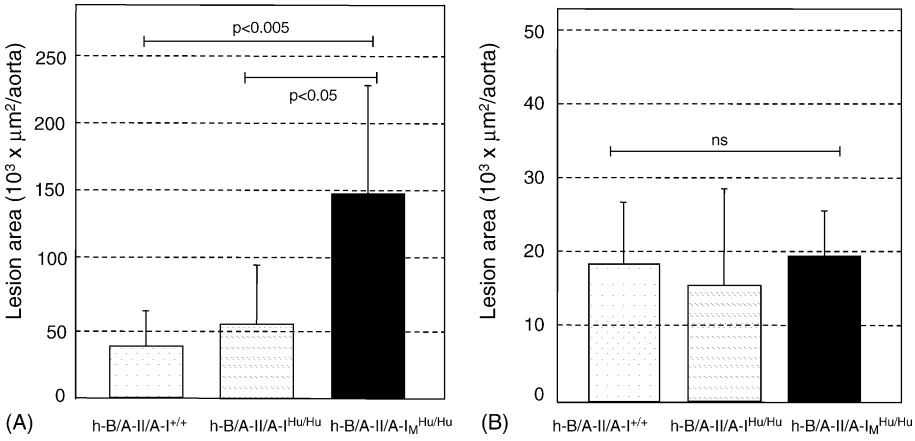



Fig. 3. Lesion area development after atherogenic diet. Lesion areas in h-B/A-II/A-I$^{+/+}$, h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$_M$$^{Hu/Hu}$ transgenic lines were determined by measuring the oil-red O stained lesions as described in Section 2. The bar graph represents the mean ± S.D. from 10 female (A) or 10 male (B) mice of each genotype.

As observed for female mice, cholesterol and triglyceride distribution profiles of pooled plasma samples from males were consistent with the lipid data shown in Table 3 (data not shown).

### 3.4. Lesion development

After 18 weeks of the atherogenic diet, mice were sacrificed and their hearts were collected for histological analysis. Consistent with previous studies [19,20], the dietary treatment resulted in relatively large fatty streak lesions in h-B/A-II/A-I$^{+/+}$ mice (Fig. 3). In this model, the mean lesion area detected in h-B/A-II/A-I$^{+/+}$ females was not different from that measured in male mice. In females, the effect of human apoA-I$_M$ expression on diet-induced atherogenesis differed significantly from that of the human *apoA-I* or mouse *apoA-I* gene: the mean lesion area in the h-B/A-II/A-I$_M$$^{Hu/Hu}$ line was greater than that observed in h-B/A-II/A-I$^{Hu/Hu}$ and h-B/A-II/A-I$^{+/+}$ mice (Fig. 3A). On the contrary, in male mice, the expression of either human apoA-I (h-B/A-II/A-I$^{Hu/Hu}$) or apoA-I$_M$ (h-B/A-II/A-I$_M$$^{Hu/Hu}$) did not affect the extent of lesion formation which was similar to that measured in the h-B/A-II/A-I$^{+/+}$ line (Fig. 3B).

## 4. Discussion

Despite numerous studies involving administration or genetic expression of apoA-I or apoA-I$_M$, no in vivo studies have directly compared the efficacy of the two apolipoproteins in atherosclerosis regression/prevention. To address the question whether apoA-I$_M$ is more efficacious than wild-type apoA-I in protecting from atherosclerosis development, previously generated k-in mice for human apoA-I or apoA-I$_M$ [12] were used for the study. The replacement of the endogenous mouse coding regions with the equivalent region of the human gene, i.e. the generation of a k-in model, is the most appropriate approach for comparison of different transgenic lines. Moreover, genetically modified mice seem to adequately reproduce the major features observed in human carriers, since apoA-I$_M$ transgenic mice display a severe hypoalphalipoproteinemia [18,12]. The two k-in lines were

crossed with mice expressing human apoB/A-II, susceptible to diet-induced atherosclerosis.

On a chow diet, while the presence of the human apoA-I transgene in the apoB/A-II background did not alter the lipid/lipoprotein profile of apoB/A-II/A-$I^{+/+}$ transgenic mice, apoA-$I_M$ expression exacerbated the pro-atherogenic profile of apoB/A-II/A-$I^{+/+}$ mice. h-B/A-II/A-$I_M^{Hu/Hu}$ mice displayed a marked hypoalphacholesterolemia, a more severe accumulation of non-HDL cholesterol and a significant increase in triglyceride levels (Table 1, Fig. 1).

No differences in apoA-$I_M$ levels were observed between previously generated human apoA-$I_M$ k-in [12] and h-B/A-II/A-$I_M^{Hu/Hu}$ mice ($100.68 \pm 19.43$ in apoA-$I_M$ k-in versus $109.66 \pm 30.79$ in h-B/A-II/A-$I_M^{Hu/Hu}$ mice). On the contrary, human apoA-I plasma concentration in the h-B/A-II/A-$I^{Hu/Hu}$ line was ∼50% of that measured in our previously reported human apoA-I k-in mice [12] ($112.75 \pm 24.29$ in h-B/A-II/A-$I^{Hu/Hu}$ versus $213.70 \pm 41.70$ in apoA-I k-in). The reason for this is unknown, but it may reflect changes in intracellular processing of human apoA-I in the FVB strain and/or may be related to the presence of the other transgenes that may affect human apoA-I metabolism in an undetermined fashion.

The extent of atherosclerosis on chow diet was evaluated in eight mice (four females and four males), aged between 25 and 30 weeks, for each transgenic line. Consistently with previous studies performed on comparable mouse models [20,23], only very few and small atherosclerotic lesions were detected both in male and female mice (the mean lesion area was $<0.1 \times 10^3$ ($m^2$/aorta for each animal) and no significant differences were observed among the three transgenic lines ($p > 0.05$) (data not shown). This result suggested that in apoB/A-II mice a pro-atherogenic diet is required to induce lesion development; thus, a commonly used high-fat, cholate-containing diet was chosen to induce atherosclerosis. This dietary approach may have some limitations, since cholate, besides inducing hypercholesterolemia [24], determines other unwanted effects in lipid metabolism [21]. However, when a high fat diet is administered to mice in the absence of cholate, lesions develop in very low extent [21].

As expected, this atherogenic diet strongly modified the lipid/apolipoprotein profile of the transgenic lines analyzed. Males and females of both h-B/A-II/A-$I^{Hu/Hu}$ and h-B/A-II/A-$I_M^{Hu/Hu}$ lines responded differently to the diet. In fact, females of both lines displayed a more dramatic elevation of total cholesterol compared to males. Similarly to what observed on chow diet, in h-B/A-II/A-$I_M^{Hu/Hu}$ mice a more atherogenic profile was seen, compared to apoA-I expressing mice. Cholesterol increase was in fact particularly pronounced in h-B/A-II/A-$I_M^{Hu/Hu}$ females compared to h-B/A-II/A-$I^{Hu/Hu}$ females. In h-B/A-II/A-$I_M^{Hu/Hu}$ females more elevated triglyceride and apoB levels were also observed, although, because of some variability within the transgenic lines, a statistical significance was not achieved. This variability may be consequent to the long experimental period and to the non-homogeneous genetic background of the mice used in this study. h-B/A-II/A-$I_M^{Hu/Hu}$ male mice did not display significant differences in total cholesterol and apoB levels compared to h-B/A-II/A-$I^{Hu/Hu}$ males, although these parameters were moderately increased in the former (Table 3). The mechanism by which apoA-$I_M$ exacerbates the pro-atherogenic profile seen in the other mouse lines is not clear, but it may be associated with the observation that apoA-$I_M$ overexpression in transgenic mice [18] increased triglyceride synthesis and consequently increased VLDL production [25]. This effect might be exacerbated when apoA-$I_M$ expression is associated to a genetically, and/or dietary-induced atherogenic background.

Finally, while in h-B/A-II/A-$I^{Hu/Hu}$ mice the atherogenic diet led to a reduction of HDL cholesterol and apoA-I levels, in h-B/A-II/A-$I_M^{Hu/Hu}$ mice the atherogenic diet resulted in no changes (females) or an elevation of both parameters (males); consequently, after the high fat diet HDL cholesterol levels were more elevated in apoA-$I_M$ than in apoA-I expressing mice.

Histological analysis of atherosclerotic lesions at the end of the dietary treatment showed no protection by human apoA-I over h-B/A-II/A-$I^{+/+}$ mice (Fig. 3). This apparent discrepancy, in comparison to the results of previous animal studies with apoA-I transgenic mice [26,27], can be explained by the choice of the model used for the present study, in which only one copy of the human apolipoprotein is expressed at both alleles in place of the murine counterpart. Other apoA-I transgenic models possessed multiple copies of the *apoA-I* gene, and the sites of integration are unknown. Based on our results, the effect of human apoA-I in atherosclerosis progression appears to be comparable to that of murine apoA-I, when a single copy of the two genes is integrated into the mouse genome.

No difference between h-B/A-II/A-$I_M^{Hu/Hu}$ and h-B/A-II/A-$I^{Hu/Hu}$ males was detected in atherosclerotic lesion progression suggesting no protective advantage of the *apoA-$I_M$* gene over that of the *apoA-I* gene. Our results on male mice support the clinical observation on apoA-$I_M$ carriers, who have a degree of carotid atherosclerosis similar to matched related controls [4]. Differently from males, apoA-$I_M$ expressing females developed greater atherosclerotic lesions than those in h-B/A-II/A-$I^{Hu/Hu}$ female mice, possibly as a consequence of their pro-atherogenic profile, characterized by higher total cholesterol and apoB plasma levels. The elevated HDL plasma concentrations observed in apoA-$I_M$ mice after the pro-atherogenic diet did not seem to play a major role in protecting from arterial lesion development. Although speculative, this may be due to an accumulation of dysfunctional HDL that could impair reverse cholesterol transport. Similar observations have been reported for other genetically modified mice that had elevated HDL without arterial lesion protection [28,29].

From these results, it can be concluded that in our genetically modified mouse model, the expression of apoA-$I_M$ may be as protective as that human apoA-I (i.e. h-B/A-II/A-$I_M^{Hu/Hu}$ males). However, when apoA-$I_M$ mice respond to

genetic and dietary pro-atherogenic stimuli with an exaggerated high-risk lipoprotein profile (i.e. h-B/A-II/A-$I_M$$^{Hu/Hu}$ females), a loss of protection against arterial lesion development is observed. The mechanism(s) contributing to this unexpected response is currently under investigation to better understand the impact of the apoA-$I_M$ mutant in lipoprotein metabolism and atherosclerosis protection.

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